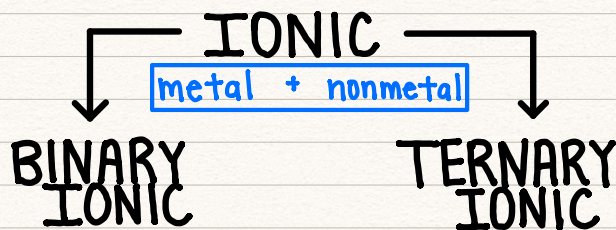


CHEMICAL NOMENCLATURE FLOWCHART



metal + nonmetal

Ex: Cation⁺ Anion⁻

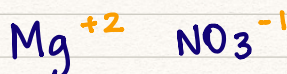


Formula = Li₂S

Name = lithium sulfide

metal + polyatomic ion

Ex: Cation Anion



Formula = Mg(NO₃)₂

Name = magnesium nitrate

Rule: Name the cation with the original name of the element, and change the ending of the anion name to ide.

Rule: Name the cation with the original name of the element, and name the anion according to the correct name for the polyatomic ion.

DIATOMIC MOLECULES

Single element that travels in a pair

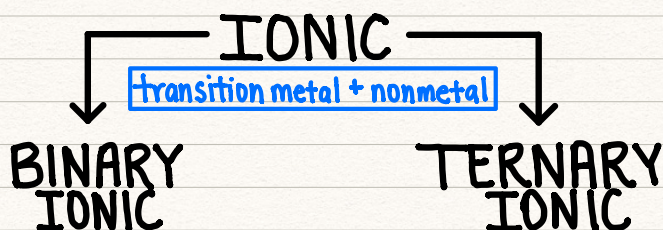
"Lucky 7"

7 elements: Starts with #7, looks like a 7

* N₂, O₂, F₂, Cl₂, Br₂, I₂, H₂*

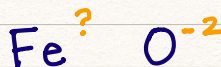
The Br-I-N-Cl-H-O-F twins always travel in pairs

* Don't Forget: You must put the polyatomic ion in parenthesis if you have more than one.



transition metal + nonmetal

Ex: Cation Anion



Formula = Fe₂O₂ $\xrightarrow{\text{reduce}}$ FeO

Name = iron (II) oxide

transition metal + polyatomic ion

Ex: Cation Anion



Formula = CuNO₃

Name = copper (I) nitrate

Rule: name the cation with the original name of the element (you must use a roman numeral next to the cation to show the charge of the transition metal), and change the ending of the anion name to ide.

Rule: name the cation with the original name of the element (you must use a roman numeral next to the cation to show the charge of the transition metal), and name the anion according to the correct polyatomic ion.

TRANSITION METAL EXCEPTIONS

transition metals that do not follow the roman numeral rule

In addition to the transition metals, lead (Pb) and tin (Sn) also create multiple charges and must have a roman numeral when naming.

Silver (Ag⁺¹), Cadmium (Cd⁺²) and Zinc (Zn⁺²) do not need roman numerals

COVALENT

nonmetal + nonmetal

BINARY MOLECULAR

nonmetal + nonmetal



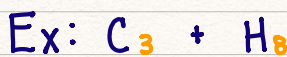
* no charges *

Formula: NO_2

Name: nitrogen dioxide

HYDROCARBONS (FUELS)

Carbon + hydrogen



* no charges *

Formula: C_3H_8

Name: propane

Rule: use the binary molecular prefixes to show the amount of each element present, and change the ending of the second element to ide.
* never use the prefix mono on the first element *

Rule: use the hydrocarbon prefixes to show the amount of carbon in the molecule and end the name in ane to show that it is an alkane.

BINARY MOLECULAR PREFIXES

mono = 1
di = 2
tri = 3
tetra = 4
penta = 5
hexa = 6
hepta = 7
octa = 8
nona = 9
deca = 10

HYDROCARBON PREFIXES

meth = 1
eth = 2
prop = 3
but = 4
pent = 5
hex = 6
hept = 7
oct = 8
nona = 9
deca = 10

hydrocarbons come in several types depending on the number of bonds in their structure

Alkanes = single bond

Alkenes = double bond

Alkynes = triple bond

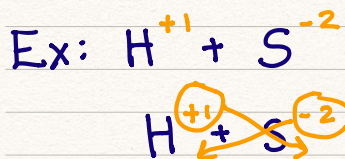
* we will only be naming alkanes *

ACIDS & BASES

ACIDS

BINARY ACIDS

H + nonmetal

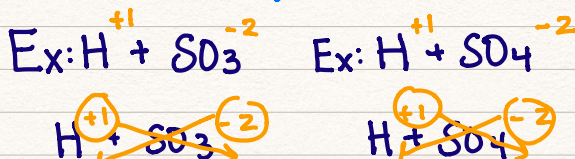


Formula: H_2S

Name: hydrosulfuric acid

TERNARY ACIDS

H + polyatomic ion



Formula: H_2SO_3

Name: sulfurous acid

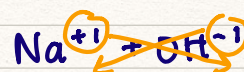
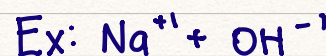
Formula: H_2SO_4

Name: sulfuric acid

BASES

HYDROXIDES

ending in OH



Formula: $NaOH$

Name: sodium hydroxide

Rule: Use the prefix hydro and change the ending of the anion to ic to create one word. Write the word acid at the end.

Rule: Do not use the prefix hydro! Write the name of the polyatomic ion and change the ending to ic if the ion ends in ate and ous if the ion ends in ite.

Rule: Name the cation with the original name of the element and name the anion as hydroxide.

POLYATOMIC IONS

Acetate:	$C_2H_3O_2$	1^-
Ammonium:	NH_4	1^+
Arsenate:	AsO_4	3^-
Bromate:	BrO_3	1^-
Carbonate:	CO_3	2^-
Chlorate:	ClO_3	1^-
Chlorite:	ClO_2	1^-
Chromate:	CrO_4	2^-
Cyanide:	CN	1^-
Dichromate:	Cr_2O_7	2^-
Hydrogen Carbonate (bicarbonate):	HCO_3	1^-
Hydrogen Sulfate (bisulfate):	HSO_4	1^-
Hydrogen Sulfite (bisulfite):	HSO_3	1^-
Hydroxide:	OH	1^-
Hypochlorite:	ClO	1^-
Iodate:	IO_3	1^-
Nitrate:	NO_3	1^-
Nitrite:	NO_2	1^-
Oxalate:	C_2O_4	2^-
Perchlorate:	ClO_4	1^-
Permanganate:	MnO_4	1^-
Phosphate:	PO_4	3^-
Phosphite:	PO_3	3^-
Sulfate:	SO_4	2^-
Sulfite:	SO_3	2^-

per = one more oxygen than -ate

ite = one less oxygen than -ate

hypo = one less oxygen than -ite

PERIODIC TABLE OF ELEMENTS

Charges

+1 +2

(Multi - Charge)

+3 +4 -3 -2 -1 $\emptyset$

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18

1

H

Hydrogen

1.008

3

Li

Lithium

6.94

11

Na

Sodium

22.990

19

K

Potassium

39.098

37

Rb

Rubidium

85.468

55

Cs

Caesium

132.91

87

Fr

Francium

(223)

4

Be

Beryllium

9.0122

12

Mg

Magnesium

24.305

20

Ca

Calcium

40.078

38

Sr

Strontium

87.62

56

Ba

Barium

137.33

88

Ra

Radium

(226)

13

Al

Aluminium

26.982

21

Sc

Scandium

44.956

39

Y

Yttrium

88.906

57-71

89-103

14

Si

Silicon

28.085

22

Ti

Titanium

47.867

40

Zr

Zirconium

91.224

72

Hf

Hafnium

178.49

104

Rf

Rutherfordium

(267)

15

P

Phosphorus

30.974

23

V

Vanadium

50.942

41

Nb

Niobium

92.906

73

Ta

Tantalum

180.95

105

Db

Dubnium

(268)

16

S

Sulfur

32.06

24

Cr

Chromium

51.996

42

Mo

Molybdenum

95.95

74

W

Tungsten

183.84

106

Sg

Seaborgium

(269)

17

Cl

Chlorine

35.45

25

Mn

Manganese

54.938

43

Tc

Technetium

(98)

75

Re

Rhenium

186.21

107

Bh

Bohrium

(270)

18

Ar

Argon

39.948

26

Fe

Iron

55.845

44

Ru

Ruthenium

101.07

76

Os

Osmium

190.23

108

Hs

Hassium

(277)

27

Co

Cobalt

58.933

45

Rh

Rhodium

102.91

77

Ir

Iridium

192.22

109

Mt

Meitnerium

(278)

28

Ni

Nickel

58.693

46

Pd

Palladium

106.42

78

Pt

Platinum

195.08

110

Ds

Darmstadtium

(281)

29

Cu

Copper

63.546

47

Ag

Silver

107.87

79

Au

Gold

196.97

111

Rg

Roentgenium

(282)

30

Zn

Zinc

65.38

48

Cd

Cadmium

112.41

80

Hg

Mercury

200.59

112

Cn

Copernicium

(285)

31

Ga

Gallium

69.723

49

In

Indium

114.82

81

Tl

Thallium

204.38

113

Nh

Nihonium

(286)

32

Ge

Germanium

72.630

50

Sn

Tin

118.71

82

Pb

Lead

207.2

114

Fl

Flerovium

(289)

33

As

Arsenic

74.922

51

Sb

Antimony

121.76

83

Bi

Bismuth

208.98

115

Mc

Moscovium

(290)

34

Se

Selenium

78.971

52

Te

Tellurium

127.60

84

Po

Polonium

(209)

116

Lv

Livermorium

(293)

35

Br

Bromine

79.904

53

I

Iodine

126.90

85

At

Astatine

(210)

117

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Tennessine

(294)

36

Kr

Krypton

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Xenon

131.29

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.



Ptable.com

57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)